FIITJEE

ALL INDIA TEST SERIES

PART TEST – I

JEE (Main)-2019-20

TEST DATE: 9-11-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

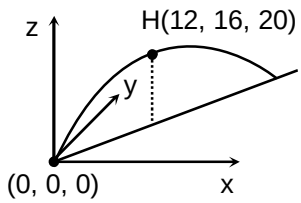
1. B

Sol. Particle will be at H_{max} at t = 2

$H_{max} = \frac{20 \times 20}{2 \times 10} = 20m$

$\vec{r}_H = 12\hat{i} + 16\hat{j} + 20\hat{k};\ |\vec{r}| = 20\sqrt{2}$

$\vec{v}_H = 6\hat{i} + 8\hat{j}$



$\vec{\omega} = \frac{\vec{r} \times \vec{v}}{|\vec{r}|^2} = \frac{-160\hat{i} + 120\hat{j}}{\left(20\sqrt{2}\right)^2} = -\frac{1}{5}\hat{i} + \frac{3}{20}\hat{j}$

2. B

Sol. Force when applied parallel to plane will have minimum magnitude.

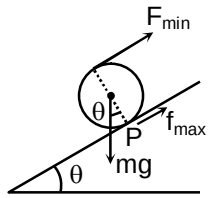
Torque about P;

$F(2r) = mg\sin\theta \times r$

$\therefore\ F_{min} = \frac{mg\sin\theta}{2}$ ....(i)

Also, $F + f_{max} = mg\sin\theta$ ....(ii)

$\therefore\ \mu = \frac{\tan\theta}{2} = \frac{3}{8}$



3. C

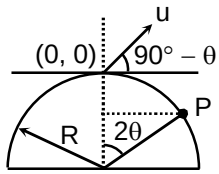
Sol. Coordinates of P [(R sin 2 θ, – R (1 – cos 2θ)]

$$y = x\tan(90-\theta) - \frac{1}{2}g\frac{x^2}{\left[u\cos(90-\theta)\right]^2}$$

$$-R\left[1-\cos 2\theta\right] = R\sin 2\theta\frac{\cos\theta}{\sin\theta} - \frac{1}{2}g\frac{R^2\sin^2 2\theta}{u^2\sin^2\theta}$$

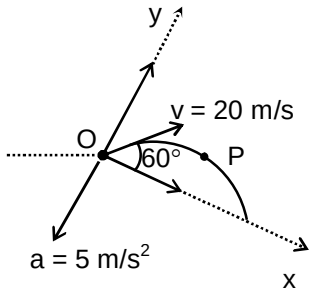
$$-2\sin^2\theta = 2\cos^2\theta - \frac{1}{2}g\frac{R4\sin^2\theta\cos^2\theta}{u^2\sin^2\theta}$$

$$\frac{2gR\cos^2\theta}{u^2} = 2 \Rightarrow u = \sqrt{gR}\cos\theta$$



4. D

Sol. Particle will describe a parabolic path as shown
After long time $R_{max} \to \infty$
$\therefore \frac{R_{min}}{R_{max}} = 0.$



5. B

Sol. Till first sand particle touches lower compartment of hourglass, weight will decrease, then it becomes constant, when last sand particle leave upper compartment it increases.

6. C

Sol. $a_Q > a_P$

$$\frac{3g\sin\theta r}{\ell} > \frac{4}{3}g\sin\theta$$

$$\Rightarrow r > \frac{4\ell}{9}$$

7. A

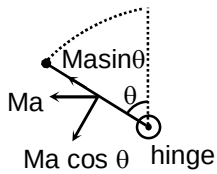
Sol. MI of door about hinge : $\frac{Mw^2}{3}$

Using : $\tau = I\alpha$

$$Ma\left(\frac{w}{2}\right)\cos\theta = \frac{Mw^2}{3}\alpha$$

$$\alpha = \frac{3}{2}\frac{a}{w}\cos\theta$$

$$\omega\frac{d\omega}{d\theta} = \frac{3}{2}\frac{a}{w}\cos\theta$$



$\Rightarrow \omega^2 = \dfrac{3a}{w}\sin\theta$

For circular motion of CM of door;

$R_H - Ma\sin\theta = M\omega^2\left(\dfrac{w}{2}\right)$

$\therefore R_H = Ma\sin\theta + \dfrac{3}{2}Ma\sin\theta = \dfrac{5}{2}Ma\sin\theta = \dfrac{125}{2}\sin\theta$

8. C

Sol. Use the definition of kinematics variables.

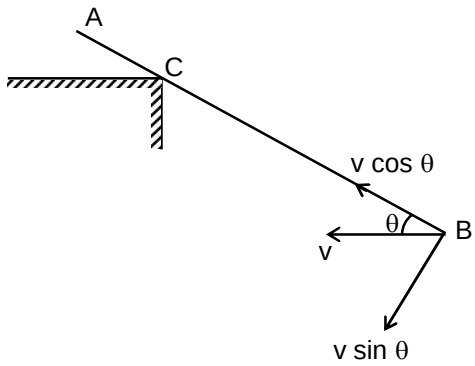
9. A

Sol. $\vec{v}_{insect} = (u - gt)\hat{j}$, $v = 0$ at $t = \dfrac{u}{g}$

$\therefore\ y_{max^2} = \displaystyle\int_0^{u/g} vdt = \int_0^{u/g}(u - gt)dt = \frac{u^2}{2g}$

10. B

Sol. $\omega_{A/B} = \omega_{C/B} = \dfrac{v\sin\theta}{CB} = \dfrac{\omega_0 rh}{(x^2 + h^2)}$



11. B

Sol. $mv_0 = (M + m)v'$

$v' = \dfrac{mv_0}{M + m}$

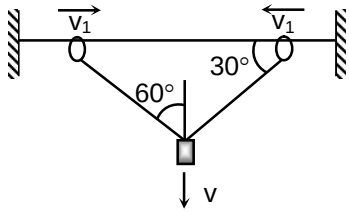
$t = t_0 + \dfrac{L(M + m)}{mv_0}$

12. D

Sol. $2mg\dfrac{\ell}{2} = \dfrac{1}{2}(2m)v^2 + 2\times\dfrac{1}{2}mv_1^2$ …(i)

$v_1\cos 30° = v\cos 60°$ …(ii)

Solving $v = \sqrt{\dfrac{3g\ell}{4}}$



13. B

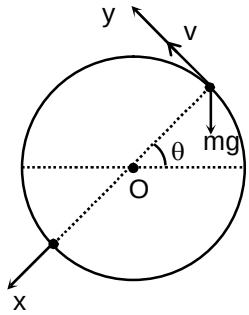
Sol. $v = \omega R$ …(i)

$mg \sin\theta = \frac{mv^2}{R}$ …(ii)

for motion after separation

$T = \frac{2V}{g\cos\theta}$ …(iii)

$2R = \frac{1}{2}g\sin\theta\left(\frac{2V}{g\cos\theta}\right)^2$ …(iv)



$gR\frac{\cos^2\theta}{\sin\theta} = Rg\sin\theta \Rightarrow \theta = 45°$

$\therefore \omega^2 R = \frac{g}{\sqrt{2}}$ or $\omega = \sqrt{\frac{g}{\sqrt{2}R}}$

14. B

Sol. $\vec{v}(t) = 2\hat{i} + (4t)\hat{j}$

$\therefore \tan\theta = 2t$

$\frac{d\theta}{dt} = \frac{2}{\sec^2\theta} = \frac{2}{17}\text{rad/sec}$ (at $t = 2\,\text{sec}$)

15. B

Sol. Centre of mass of wedge and ball will not shift in x-y plane

Let wedge displacement be $x\hat{i} + y\hat{j}$

$\vec{S}_r = (-R\cos\theta)\hat{i} + R\hat{j}$

$\therefore\ m\left[(x - R\cos\theta)\hat{i} + (y + R)\hat{j} + M(x\hat{i} + y\hat{j})\right] = 0$

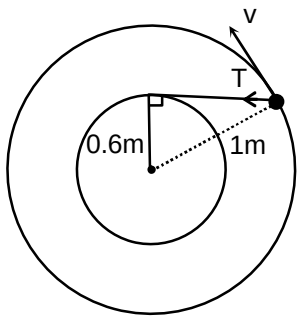
$x = \frac{mR\cos\theta}{m+M}$; $y = -\frac{mR}{m+M}$

$\vec{S}_w = \frac{1}{5}\hat{i} - \frac{2}{5}\hat{j}$

16. C

Sol. $T\cos\theta = \frac{mv^2}{R}$

$\Rightarrow\ v = \sqrt{\frac{TR\cos\theta}{m}} = 10\,\text{m/s}$



17. A

Sol. $a = \frac{F_0}{m}\cos\omega t$

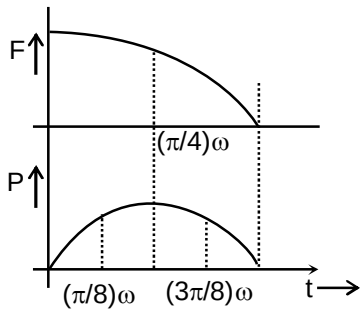
$v = \frac{F_0}{m\omega}\sin\omega t$

$\therefore$ peak power at $t = \frac{\pi}{4\omega}$.

Ratio of power is 1

Also $\int_0^{\pi/2\omega} (F - mg)dt = mv$

$v = \frac{F_0}{m\omega} - \frac{g\pi}{2\omega}$

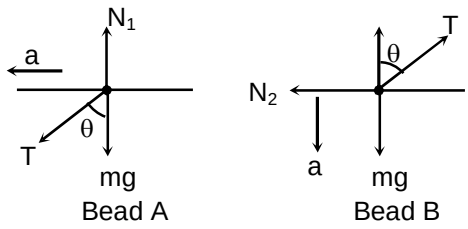


18. D

Sol. $T\sin\theta = ma$ …(i)

$mg - T\cos\theta = ma$

$T = \frac{mg}{\sin\theta + \cos\theta} = \frac{mg}{\sqrt{2}}$ ($\theta = 45^o$)



19. B

Sol. $W_{ext} = \int \vec{F}_{ext} d\vec{r} = \int_{r_i}^{r_f} \left(\frac{2\alpha}{r^3} - \frac{\beta}{r^2}\right) dr$

$= \left[-\frac{\alpha}{r^2} + \frac{\beta}{r}\right]_{2r_0}^{r_0}$

$W_{ext} = -\frac{3\alpha}{4r_0^2} + \frac{\beta}{2r_0}$

Also, at equilibrium; $F = 0 \Rightarrow r_0 = \frac{2\alpha}{\beta}$

$\therefore\ W_{ext} = \frac{\beta^2}{4\alpha} - \frac{3\alpha\beta^2}{4\times 4^{a^2}} = \frac{\beta^2}{16\alpha}$

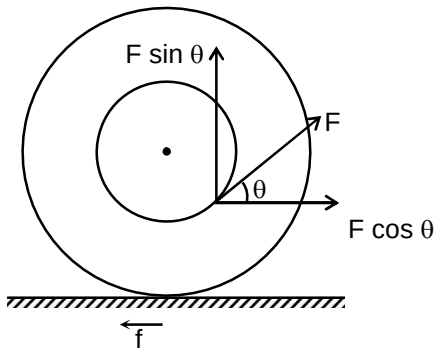
20. A

Sol. $F\cos\theta - f = ma_C$ …(ii)

$f(2R) - FR = 2mR^2\alpha$ …(ii)

$a_C = \alpha 2R$ …(iii)

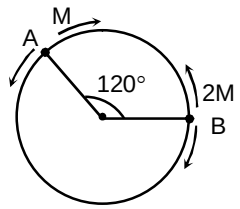
$\therefore f = \frac{F}{3}(1+\cos\theta) \Rightarrow$ always leftward.



SECTION – B

21. 5

Sol. A and B can swipe their position by walking clockwise / anticlockwise



For $\Delta\theta_{max}$: A should turn 120° (clockwise) while B should turn 240°(clockwise)

Conserving angular momentum about C:

$$MR^2(120^\circ-\theta)+2MR^2(240^\circ-\theta)+\frac{4MR^2}{2}(-\theta)=0$$

$\Rightarrow \theta = 120^\circ$

For $\Delta\theta_{min}$: A should turn 120° clockwise, B should turn 120° anticlockwise

$$\theta_{min}=\frac{120^\circ}{5}$$

$$\therefore \frac{\theta_{max}}{\theta_{min}}=5$$

22. 1

Sol. For t_{min}, striker should be shot perpendicular to line AB

$$AB=\sqrt{(3-7)^2+(4-1)^2}=5\text{ m}$$

$$\therefore \frac{1}{2}\times h\times AB=\frac{1}{2}|\overrightarrow{OA}\times\overrightarrow{OB}|$$

$$\frac{5h}{2}=\frac{1}{2}\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\3&4&0\\7&1&0\end{vmatrix}$$

$\Rightarrow h = 5$ m

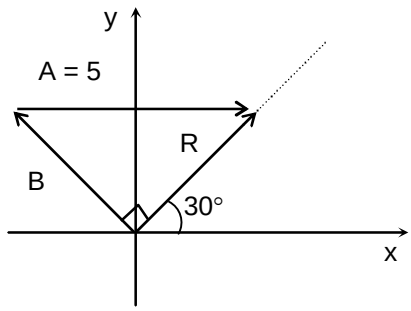
$\therefore t_{min} = 5/5 = 1$ sec

SECTION – C

23. 00004.33

Sol. From triangle;

$$R = 5\cos 30^\circ = \frac{5\sqrt{3}}{2}$$



24. 00006.80

Sol. $f_{max} = 0.25 \times 4 \times 10 = 10$ N

Net impulse is zero in 11 sec

$\Rightarrow I_f = I_F$

$\Rightarrow \frac{1}{2} \times 8 \times F_{max} = 100$

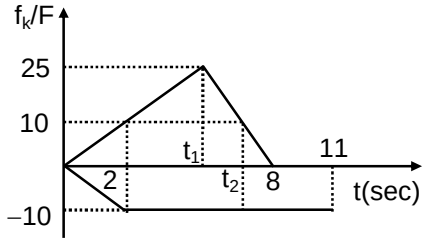
$F_{max} = 25$N

From similar triangle

$\frac{25}{t_1} = \frac{10}{2} \Rightarrow t_1 = 5\,\text{sec}$

Block will acquire V_{max} at $t = t_2$ where

$\frac{25}{3} = \frac{10}{(8 - t_2)} \Rightarrow t_2 = 6.8\,\text{sec}$



25. 00001.35

Sol. Given : $\frac{2\pi}{2\pi}(1 - \cos\theta) = 0.2$

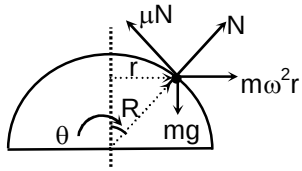
$\Rightarrow \cos\theta = 0.8$

From FBD of dust particle:

$N = mg\cos\theta - m\omega^2 r\sin\theta$ …(i)

$\mu N = mg\sin\theta + m\omega^2 r\cos\theta$ …(ii)

$\therefore\ \mu = \frac{g\sin\theta + \omega^2 r\cos\theta}{g\cos\theta - \omega^2 r\sin\theta} = 1.35$



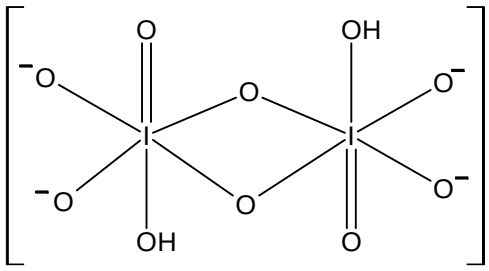
Chemistry

PART – II

SECTION – A

26. A

Sol.



27. C

Sol. $Zn = 906\ kJ\ mol^{-1}$

$Ga = 579\ kJ\ mol^{-1}$

$Ge = 761\ kJ\ mol^{-1}$

$As = 947\ kJ\ mol^{-1}$

28. D

Sol. $C_2(12) \rightarrow \text{actual} \rightarrow \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 = \pi 2p_y^2$

Diamagnetic

If s – p is not operational.

$\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^1 = \pi 2p_y^1$

paramagnetic

29. D

Sol. $A(g) + 2B(g) \rightleftharpoons AB_2(g)$

$$K_c = \frac{1}{2} = \frac{[AB_2]}{[A][B]^2}$$

$$\Rightarrow \frac{1}{2} = \frac{[AB_2]}{2(2)^2} \Rightarrow AB_2 = 4 \text{ moles}$$

Now, B = 4 moles

Let y moles of AB_2 is added.

$$\frac{1}{2} = \frac{(4+y)}{2\times(4)^2}$$

$$\therefore y = 12$$

30. D

31. C

Sol. $\frac{1}{2}H_2 \longrightarrow H^+ + e^-$

Takes B.E. for dissociation to H and then oxidizes slow.
Fact by NCERT.

32. A

Sol. $\Delta n_g = 0$ in I and III.

$\therefore$ Same moles at equilibrium as before.

33. D

Sol. $Na + Liq.\ NH_3 \longrightarrow \underset{(x+y)}{\left[Na(NH_3)_x\right]^+} + \left[e(NH_3)_x\right]^-$

Blue coloured and paramagnetic in nature.

34. C

Sol. O = C = C = C = O
→ Linear
→ Optically inactive
→ $4\sigma, 4\pi$ and 4 lone pair.

35. C

Sol. ΔE.N. , more polarity.

36. B

37. A

Sol. If $T_2 > T_1$
$\therefore$ At high E → Intensity increases and λ decreases.
(Reference – NCERT)

38. C

Sol. $S_8(s) + 12\bar{O}H(aq) \longrightarrow \underset{(-2)}{4S^{-2}(aq)} + \underset{(+2)}{2S_2O_3^{-2}(aq)} + 6H_2O(\ell)$

39. B

Sol. Both d-orbitals have axial overlapping in same phase forming σ bonds.

40. C

Sol. For exothermic reaction, increase in temperature decreases yield.

41. D

42. C

Sol. Al reacts with both acid and base to liberate H_2.

43. D

Sol. (A) $\frac{r^+Cs^+}{r^+Cl^-} = 0.732$.

(B) High temperature decreases coordination temperature.

(C) LiCl has pink colour.

44. D

45. C

Sol. $\Delta e_g H \rightarrow O \rightarrow -141$

$S \rightarrow -200$

$Se \rightarrow -195$

$\therefore S > Se > O$

SECTION – B

46. 1

Sol. Double dumb shell confirms the graph touches x-axis only once.
$\therefore$ 1 node.

47. 7

Sol. $4Ca_5(PO_4)_3F + 18SiO_2 + 30C \longrightarrow 3P_4 + 2CaF_2 + 18CaSiO_3 + 30CO$

SECTION – C

48. 00006.67

Sol. After mixing [AgCN] = 0.01 M, [HCN] = 0.01 M

$HCN \rightleftharpoons H^+ + CN^-$ K_a

$Ag^+ + CN^- \rightleftharpoons AgCN(s)$ $1/K_{sp}$

$Ag^+ + HCN \rightleftharpoons H^+ + AgCN(s)$ $K = \frac{K_a}{K_{sp}} = 2.25 \times 10^6$

0.01	0.01	–	–
x	x	0.01 – x	0.01 – x

$K = \frac{0.01 - x}{x^2}$ since x is small w.r.t. 0.01

$$K = \frac{0.01}{x^2}$$

$$\therefore x = 6.6 \times 10^{-5} \quad M = \left[Ag^{+} \right]$$

49. 00000.71

Sol. Nearest in BCC $\frac{\sqrt{3}a}{2}$; Next to next nearest $= \sqrt{\frac{3}{2}}a$ in FCC

$$\therefore \text{Ratio} = \frac{\sqrt{3}a \times \sqrt{2}}{2 \times \sqrt{3}a} = \frac{1}{\sqrt{2}} = 0.71$$

50. 00001.30

Sol. (Oxalic) $N_1V_1 = N_2V_2$ ($KMnO_4$)

$M \times 40 \times 2 = 0.05 \times 16 \times 5$

$$M = \frac{0.05 \times 16 \times 5}{40 \times 2} = 0.05$$

pH = - log (0.05)

=1.3

Mathematics

PART – III

SECTION – A

51. B

Sol. $\log\beta = \int_0^1 \log(1+x^3)\,dx = \log 2 - 3 + \int_0^1 \frac{3}{1+x^3}\,dx = \log 2 - 3 + 3\lambda$

52. A

Sol. Put $x + \cos\alpha = \sin\alpha \tan\theta$

$\int \frac{x\cos\alpha + 1}{(x^2 + 2x\cos\alpha + 1)^{\frac{3}{2}}}\,dx = \int (\cot\alpha \sin\theta + \cos\theta)\,d\theta = -\cot\alpha \cos\theta + \sin\theta + c$

$= \frac{x}{\sqrt{x^2 + 2x\cos\alpha + 1}} + c$

53. D

Sol. $f(x) = \begin{cases} \frac{1}{x+1} & ; \quad 0 \le x < 1 \\ \frac{2}{x} & ; \quad 1 \le x < 2 \\ \frac{3}{x-1} & ; \quad 2 \le x < \frac{5}{2} \end{cases}$ f(x) is discontinuous and bijective function

$\lim_{x\to 1^-} f(x) = \frac{1}{2}$; $\lim_{x\to 1^+} f(x) = 2$ and f(1) = 2

54. D

Sol. $f(2a^2 + a + 1) < f(3a^2 - 4a + 1)$

$\Rightarrow 2a^2 + a + 1 > 3a^2 - 4a + 1$

$\Rightarrow a \in (0, 5)$

Also, $2a^2 + a + 1 > 0$ and $3a^2 - 4a + 1 > 0$, we get $a < \frac{1}{3}$ or $a > 1$

So, $a \in \left(0, \frac{1}{3}\right) \cup (1, 5)$

55. C

Sol. $f(xy + 1) = f(x)\,f(y) - f(y) - x + 2$ ..... (1)

$f(xy + 1) = f(x)\,f(y) - f(x)\,y + 2$ ..... (2)

Thus, $f(x) + y = f(y) + x$

Put $y = 0$, we get $f(x) = x + 1$

56. C

Sol. Let the point of contact be $(\alpha, \cos\alpha)$, then $\frac{y - \cos\alpha}{x - \alpha} = -\sin\alpha$

$\Rightarrow -\cot\alpha = \alpha$, then, we get $a = 1$

57. D

Sol. $1 + a + b = 0$

$c - 1 = 0 \Rightarrow c = 1$

Also, $\left.\frac{dy}{dx} = c - 2x\right|_{x=1} = -1$

$\left.\frac{dy}{dx} = 2x + a\right|_{x=1} = -1$

$\Rightarrow a = -3; b = 2$

Area bounded by $y = x^2 + ax + b$ with x-axis

$$\Rightarrow \left|\int_1^2 (x^2 - 3x + 2)dx\right| = \left|\left(\frac{x^3}{3} - \frac{3x^2}{2} + 2x\right)\right|_1^2 = \frac{1}{6}$$

58. D

Sol. Let $f(x) = \int_0^x \frac{dt}{\sqrt{4 - t^2 - t^3}} - x$

$$f'(x) = \frac{1}{\sqrt{4 - x^2 - x^3}} - 1 < 0 \quad \forall\ x \in (0, 1)$$

$\Rightarrow f(x)$ is decreasing in (0, 1)

Also, $f(0) = 0$

$\Rightarrow f(x) < 0\ \forall\ x \in (0, 1)$

$\Rightarrow$ No solution

59. B

Sol. Let first x_1 elements of set A maps with b_1 and next x_2 elements maps with b_2 and so on last x_{50} elements maps with b_{50}

$\Rightarrow x_1 + x_2 + ... + x_{50} = 100$

Number of positive integral solutions = ${}^{99}C_{49}$

60. C

Sol. $\lim_{x \to 0^+} x^x = 1$

61. B

Sol. Put $y = x$, we get $2f(x) + (f(x))^2 = 1$

Differentiate w.r.t to x, we get $2f'(x) + 2f(x)f'(x) = 0$

$\Rightarrow f'(x)\ (1 + f(x)) = 0$

$\Rightarrow f'(x) = 0$, because $f(x) > 0$

62. D

Sol. $f(x) = \sum_{n=0}^{\infty} \frac{(x\log a)^n}{n!} = e^{x\log a} = a^x$

$f(x) = a^x$ always continuous and differentiable

63. A

Sol. $\cos\pi\sqrt{n^2+n} = (-1)^n \cos\left(n\pi - \pi\sqrt{n^2+n}\right)$

$\lim\limits_{n\to\infty} n - \sqrt{n^2+n} \Rightarrow -\frac{1}{2}$

$\lim\limits_{n\to\infty} \cos\left(\pi\sqrt{n^2+n}\right) = (-1)^n \cos\frac{\pi}{2} = 0$

64. B

Sol. ' / ' is reflexive since every natural number is a factor of itself
' / ' is transitive. If n is a factor of m and m is a factor of P, then surely n is a factor of P
' / ' is not symmetric, for example 2 is a factor of 4 but 4 is not a factor of 2

65. D

Sol. $f(3x) - f(x) = x$

$\Rightarrow$ Replace x by $\frac{x}{3}$, we get

$f(x) - f\left(\frac{x}{3}\right) = \frac{x}{3}$

$f\left(\frac{x}{3}\right) - f\left(\frac{x}{9}\right) = \frac{x}{9}$

.....

.....

$f\left(\frac{x}{3^{n-1}}\right) - f\left(\frac{x}{3^n}\right) = \frac{x}{3^n}$

Adding, we get $f(x) - f\left(\frac{x}{3^n}\right) = x\left(\frac{1}{3} + \frac{1}{9} + \frac{1}{27}.....\right)$

$\lim\limits_{n\to\infty} f(x) - f\left(\frac{x}{3^n}\right) = \frac{x}{2}$

$P(x) = \frac{x}{2}$

66. D

Sol. $I = \int_0^\infty \frac{dx}{(1+x^3)(1+x^2)}$ ..... (1)

Put $x = \frac{1}{t}$

$I = \int_0^\infty \frac{x^3}{(1+x^3)(1+x^2)} dx$ ..... (2)

Adding equation (1) and (2), we get

$2I = \int_0^\infty \frac{dx}{1+x^2} = \frac{\pi}{2}$

$I = \frac{\pi}{4}$

67. A

Sol. $f'(x) = \dfrac{1}{x^2 + f(x)^4} < \dfrac{1}{x^2+1}$

$\Rightarrow \int_1^x f'(x) < \int_1^x \dfrac{dx}{1+x^2}$

$\Rightarrow f(x) < \tan^{-1}x - \dfrac{\pi}{4} + f(1)$

$\Rightarrow f(x) < \tan^{-1}x + 1 - \dfrac{\pi}{4}$

68. A

Sol. $I(t) = \int_{-1}^{2}\left[x^2 + \left(\dfrac{t^2}{4} + \dfrac{1}{t}\right)x + \left(\dfrac{1}{4t^2} - \dfrac{1}{2t}\right)\right]dx = \dfrac{3t^2}{8} + \dfrac{3}{4t^2} + 3$

69. D

Sol. $\lim\limits_{x\to 0} f\left(2x^3 - x^2\right) = \lim\limits_{x\to 0^+} f(x) = 24$

$\lim\limits_{x\to 0} f\left(x^6 - x^2\right) = \lim\limits_{x\to 0^-} f(x) = 3$

$\Rightarrow \lambda = 8$

70. B

Sol. $2(F(x) - f(x)) = f^2(x)$

and $\dfrac{dF}{dx} = f(x)$

$F'(x) = f'(x)(1 + f(x))$

$\Rightarrow f'(x) = \dfrac{f(x)}{1+f(x)}$

$\lim\limits_{x\to\infty} f(x) \to \infty$

As f(x) is increasing surjective function

$\lim\limits_{x\to\infty} \dfrac{f(x)}{x} = 1$

SECTION – B

71. 3

Sol. The frequency of $a_i = 1$ is the same as that of $a_i = 0$ for $1 \le i \le n-1$ and $a_n = 1$, then

$S_n = \dfrac{1}{2} \times 2^{n-1} \times \left(\dfrac{1}{10} + \dfrac{1}{10^2} + ... + \dfrac{1}{10^{n-1}}\right) + 2^{n-1} \times \dfrac{1}{10^n}$

$\lim\limits_{n\to\infty} \dfrac{S_n}{T_n} = \lim\limits_{n\to\infty}\left[\dfrac{1}{18}\left(1 - \dfrac{1}{10^{n-1}}\right) + \dfrac{1}{10^n}\right] = \dfrac{1}{18}$

72. 6

Sol. Area bounded by $y^2 = 4x$ and $y = mx$ is $\frac{8}{3m^3}$

$\Rightarrow \left|\frac{8}{3m^3}\right| = 9 \Rightarrow m = \pm\frac{2}{3}$

SECTION – C

73. 00012.00

Sol. $f''(x) = 12x^2 + 6ax + 27 \geq 0$
$\Rightarrow a \in [-6, 6]$

74. 00004.25

Sol. $f'(x) = a - 2\sin 2x + \cos x - \sin x = a + 2(\cos x - \sin x)^2 + (\cos x - \sin x) - 2$

Minimum values of $2(\cos x - \sin x)^2 + (\cos x - \sin x) - 2$ is $-\frac{17}{8}$

So, $a - \frac{17}{8} \geq 0$ for increasing function

$\therefore a \geq \frac{17}{8}$

75. 00009.00

Sol. $\int_0^1 \left\{\frac{x^4}{9} - \left(\frac{x^2}{3} - f(x)\right)^2\right\} dx = \frac{1}{45}$

$\Rightarrow f(x) = \frac{x^2}{3} \Rightarrow \int_0^1 f(x)dx = \frac{1}{9}$